H524 Group Project Check-In: AI Divergence Experiment

Fall 2025

Dr. John Molitor — Due: See Canvas — 5% of Project Grade

Purpose: Discover how AI tools respond differently to biostatistical questions and learn to use prompt engineering to reach consensus.

The Experiment (4 Steps)

Step 1: Propose a Research Question - Choose a **broad question** about your dataset that might elicit different AI answers. Good questions ask about *method choice, assumptions, or interpretation.* Examples: "What statistical method for treatment effects in pre-post data?" or "How to handle three-group comparisons?"

Step 2: Consult Two Different AI Tools - Use the **EXACT same prompt** with two AI tools (Claude, ChatGPT, Copilot, or Gemini). Copy their full responses.

Step 3: Use Third AI as Judge - Ask a different AI: *"I asked two AI tools [your question] and got these responses: [paste both]. Rate their similarity 0-100 (0=different, 100=identical). Identify key agreements/disagreements."*

Step 4: Iterate to Increase Consensus - Try $\geq$2 modified prompts (add context, be specific, add constraints, request format). Re-score similarity each time.

Deliverable: 1-2 Page Report

Section 1: Team info (number, members, dataset)

Section 2: Your research question and why you expected divergence

Section 3: Experiment table showing $\geq$3 iterations with: Prompt text, AI 1 response summary, AI 2 response summary, Similarity score (0-100)

Section 4: Brief reflection (4-5 sentences): Did similarity increase? What prompt changes worked? What did you learn about AI variability? Which AI will you use for final project?

Grading (5 pts = 5% of Project)

Component	Pts
Question broad enough for divergence	1
Consulted 2 AIs with identical prompt	1
Used 3rd AI to score similarity	1
$\geq$2 prompt iterations documented	1
Thoughtful reflection	1
Total	**5**

Why This Matters: Your final project (30% of grade) requires comparing AI tools critically. This teaches you that AI responses vary by tool AND prompt. Mastering prompt engineering now strengthens your analysis later.